POLICY ENGINE

# An Open Firm-Level Microsimulation of the UK VAT Registration Threshold

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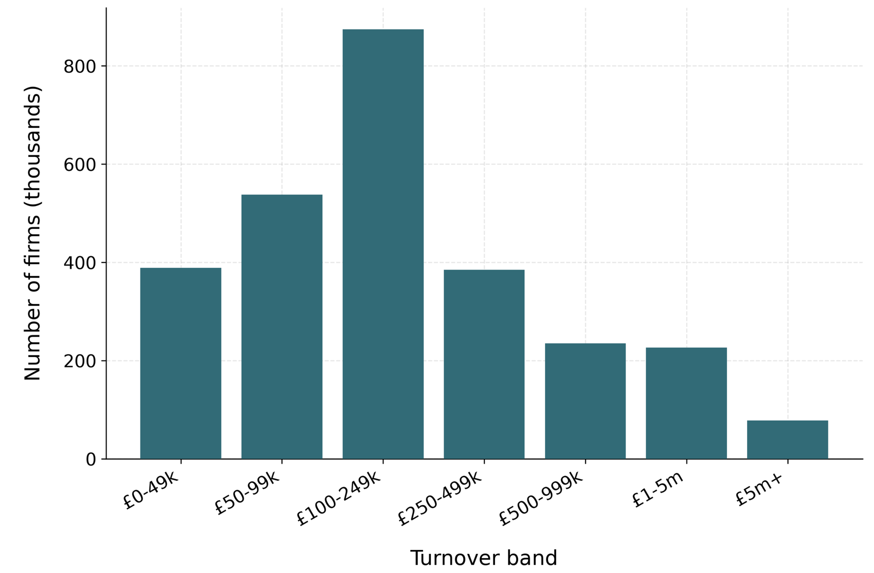
## The problem: VAT as a tax *notch*

The UK VAT registration threshold is a notch, not a kink. Once taxable turnover crosses $T^*$, 20% VAT falls due on the *whole* turnover, not just the increment — a discrete liability jump of $\tau T^* = £17{,}000$ at $T^* = £85{,}000$. Threshold £85,000 (2023–24 data year), raised to £90,000 in April 2024 — among the highest in the OECD — and it sits in a *dense* part of the firm-size distribution. Firms respond by locating just below it; the cost of crossing is sharpest for *consumer-facing* firms with little reclaimable input VAT, so the notch distorts both the size distribution and firms' growth at the margin.

*VAT/PAYE-registered businesses by turnover band — ONS, 2023–24*




## Three reform families, one basis

*What we cost.*

Costed on a *common static basis*:

- Level — move the threshold up or down.
- Shape — a graduated taper that phases VAT in.
- Rate — a reduced-rate band above the threshold.

## Contribution

**Headline: only the graduated taper removes the notch's dominated region (£550m, vs £497m for a 10% band that removes none); a level rise costs most — and has exactly zero intensive-margin behavioural offset.**

An open, reproducible firm-level microsimulation that:

1. costs level + shape + rate reforms on one static basis;
2. characterises the notch's distortion via its exact, elasticity-free dominated region;
3. proves the near-threshold shape is target-inherited (OBR-calibrated; placebo-verified) — in both directions;
4. adds a conditional behavioural layer with an exact static nesting.

## Data methodology

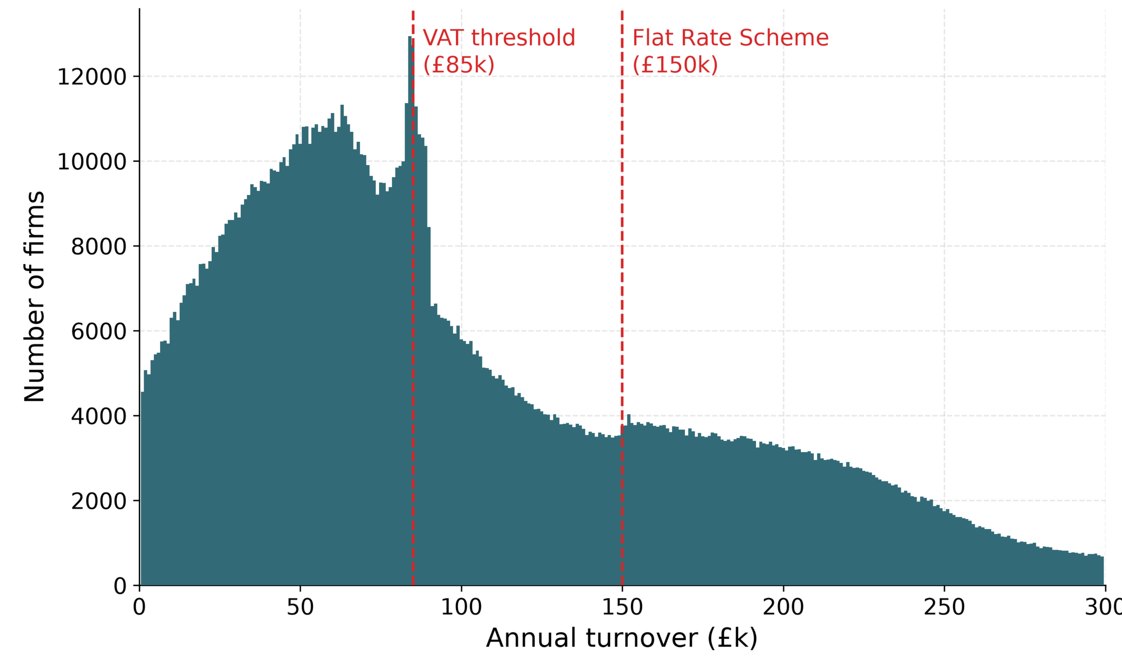
*How the synthetic data are built.*

Draw synthetic firms by sector × turnover band; give each an input share $x_i = \rho_i y_i$ and net VAT liability $v_i = 0.20\,(y_i - x_i)$ — the standard rate on value added. Firm weights $w_i = e^{\theta_i}$ are fit by gradient descent to HMRC + ONS aggregates, with the near-threshold £1k-band profile pinned to the OBR's published Chart C data. The threshold is

## Synthetic firm data

*→ The engine behind every costing.*

≈2.94m firm rows weighted to ≈2.5m UK firms, calibrated to HMRC + ONS aggregates (turnover-band counts, sector, employment, net VAT liability by band) at 89.2% / 92.8% accuracy for the 2023–24 / 2024–25 vintages. Calibrated *to* the aggregates, so agreement is an internal-consistency check, not external validation. Levels are the VAT/PAYE-registered frame: unregistered sole traders (most of the all-business cliff at £85k) are outside it.



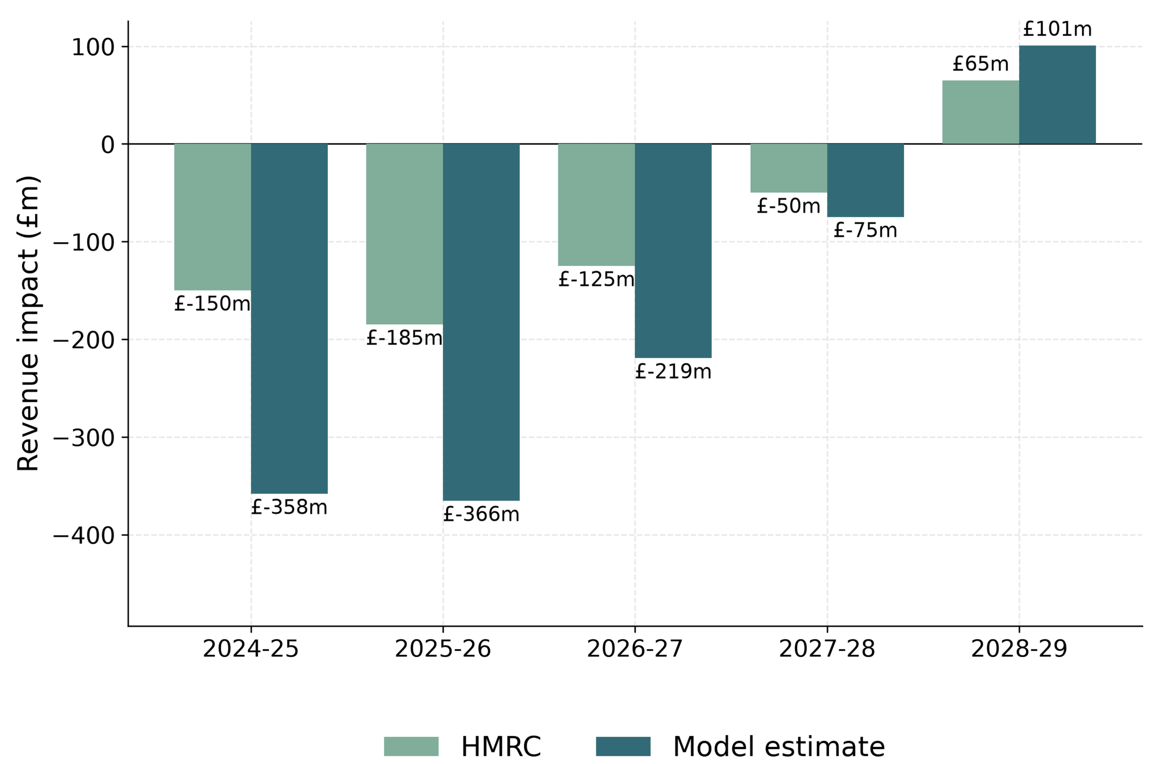

## Static costing

*Pricing a reform on this data.*

$$R(T) = \sum_i w_i\, v_i\, \mathbb{1}\{y_i \geq T\}$$

Revenue moves only through firms whose registration status flips.

Anchor vs HMRC: the £85k→£90k April-2024 rise, under two released-firm conventions, overshoots early (−£366m / −£209m retention vs published −£185m in 2025–26 — consistent with HMRC's phased registration response) and matches by 2026–27 (−£125.1m vs −£125m), with the 2028–29 sign flip reproduced.




## Threshold sweep (2025–26)

*Scaling the level reform across thresholds.*

## Reform menu

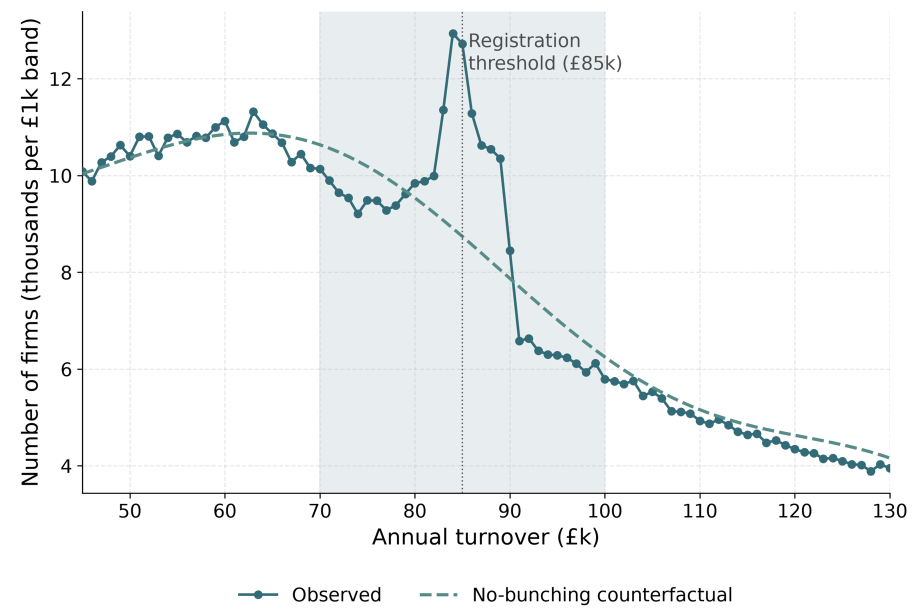
*→ Cost and distortion on a common £85k / £184.8bn base.*

| Reform | Type | Static | Dominated region |
|---|---|---|---|
| Raise to £100k | Level | −£784m | relocated |
| Taper £85–105k | Shape | −£550m | removed |
| Reduced 10% | Rate | −£497m | split in two |
| Reduced 15% | Rate | −£249m | split in two |

## The near-threshold shape is target-inherited

*A transparency result, proved in both directions.*

The 2023–24 file reproduces the administrative bunching profile because it is calibrated to it (OBR £1k-band targets); the estimator reads back $E = 7{,}873$. Regenerate *without* those targets ⇒ $E = 0$. And with *no* fine targets, the 2024–25 vintage's coarse £90k band edge alone yields $b_{\text{LLAT}} = 1.23$ — within 10% of Liu et al.'s (2021) administrative 1.361, coincidentally and unstably. Synthetic bunching is never behavioural evidence; the administrative estimate remains the behavioural fact.



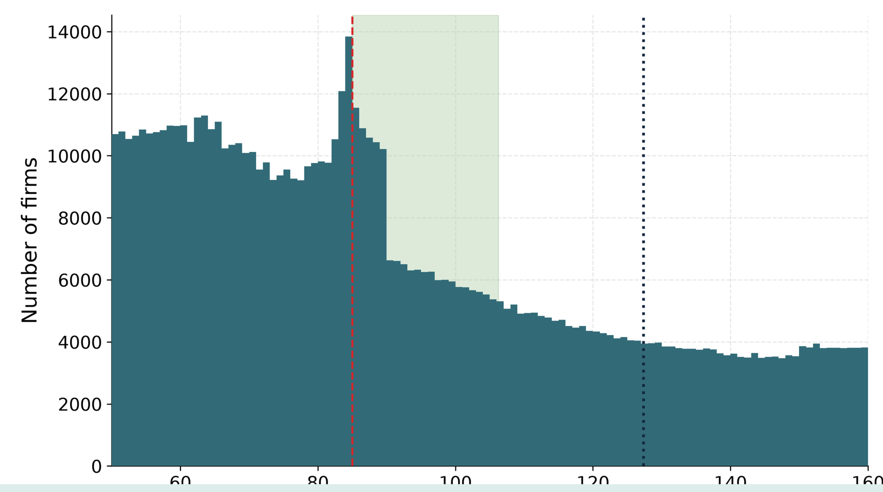

## The dominated region

*Beyond revenue: measuring the notch's distortion.*

Profit jumps *down* by $\tau T^* = £17{,}000$ at the threshold. The dominated region — turnover where no firm locates because it earns less than at $T^*$ — has exact width

$$a = T^* \tfrac{\tau}{1-\tau} = £21{,}250$$

(£85,000–£106,250), spanning ≈160,000 weighted firms. Exact, elasticity-free, and input-share invariant (the deductible share cancels in the value-added formulation).




## How reforms change it

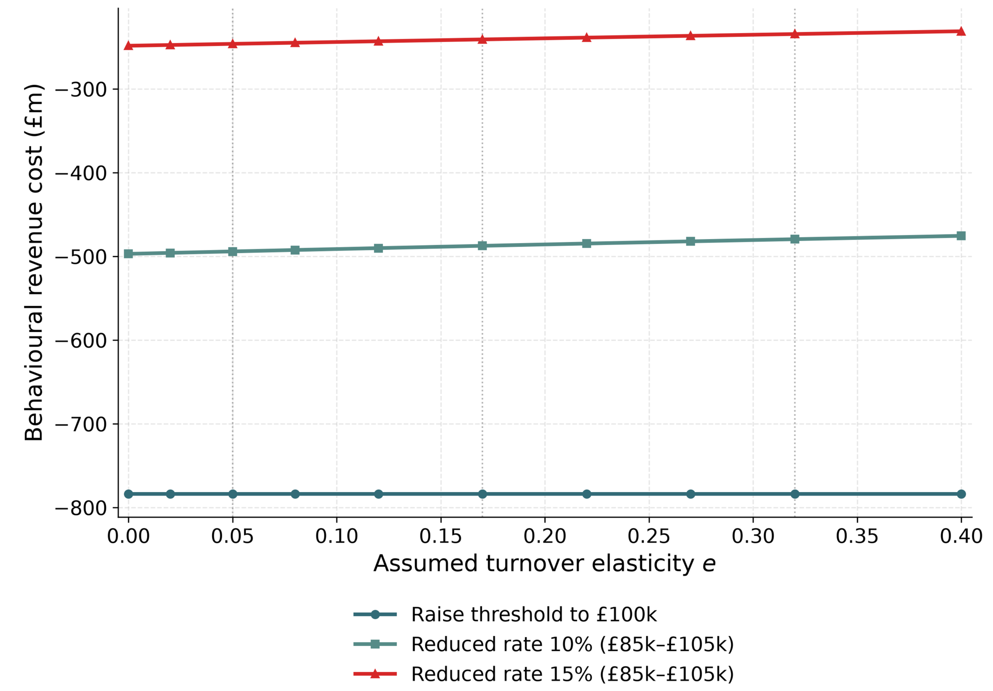
*→ Which reforms actually shrink this band?*

- Raising the threshold relocates the £21,250 band unchanged.
- A banded reduced rate does not shrink it. It lowers the *largest single* notch (£17,000→£12,750 at 15%), but reverting to 20% at the band top adds a *secondary* notch $a' = T_1 \frac{\tau - r}{1-\tau}$ of £5,250, so total width is £21,562 (15%) / £22,569 (10%) — it splits the band, not shrinks it.
- Only the graduated taper removes the distortion entirely.

## Behavioural layer

*Layering an assumed response on the static costs.*

Conditional on an assumed turnover elasticity $e$ (*not* identified on synthetic data); sweep $e \in \{0.05, 0.17, 0.32\}$. The intensive margin barely moves the costings: a level rise has exactly zero intensive-margin offset at every $e$ (released firms leave the VAT base, so their expansion is untaxed), and the reduced-rate offsets stay under 4% (10% band: £497m to £488m at $e = 0.17$). The taper is *excluded* — its rate varies continuously, outside the flat-rate machinery. Only $e = 0.05$ has an external anchor, so this is a sensitivity range, not a forecast — and the $e \to 0$ limit reproduces the static costs exactly.




## Conclusions

What this paper is: an open, reproducible microsimulation that models the UK VAT threshold as a *turnover-tax notch* on synthetic firm data, and prices level, shape and rate reforms three ways — statically, by the exact distortion each removes, and behaviourally.

1. Prices level + shape + rate on one basis; tracks HMRC's anchor costing once the deregistration convention is stated.
2. On the dominated-region metric, only the taper removes the distortion; the level move costs most and removes none of it.
3. Sub-band structure in calibrated data is target-inherited — never behavioural evidence, in either direction.
4. The intensive margin barely moves the static costings; the static cost and dominated region are the elasticity-free anchors.